SPL-II Project Proposal Report

Course Title: Software Project Lab II

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Team - 4

Project Title: UniConnect

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1 PROJECT OVERVIEW

UniConnect is a university-exclusive social networking and collaboration platform designed to connect students, alumni, clubs, and administrators within a unified digital ecosystem. Built using Flutter (mobile frontend) and Java Spring Boot (backend), the platform consolidates fragmented university communication channels into a single, purpose-built application.

1.1 Core Capabilities

- **Project Teammate Finder:** Students search for collaborators by skills and create project posts
- **Peer-to-Peer Mentorship:** Senior students offer structured, slot-based group mentorship sessions (7-10 mentees)
- **Career Board:** Alumni post job/internship opportunities; students browse by skills
- **Events & Announcements:** Clubs and admins create events with RSVP tracking
- **One-to-One Chat:** Real-time messaging between connections

1.2 Target Users

- Students (primary users)
- Alumni (job providers, recruiters)
- Club Administrators (event organizers)
- System Administrators (platform moderators)

1.3 Key Differentiators

- University-email verification (closed ecosystem)
- Skill-based discovery and matching
- Consolidated platform (replaces WhatsApp, Facebook, Email, LinkedIn)
- Mobile-first design (Flutter cross-platform)
- Graph algorithm-powered recommendations

2 MOTIVATION BEHIND THE PROJECT

2.1 The Problem: Fragmented University Communication

University students currently face significant challenges in academic networking and collaboration:

2.1.1 Communication Fragmentation

Students juggle 5-6 different platforms daily which causes important information to get lost, opportunities to be missed, and communication to be chaotic.

2.1.2 Discovery Problem: Finding the Right People

Students struggle to find collaborators with specific skills. Current solution to this is to post in random WhatsApp groups, ask friends, and hope for the best.

2.1.3 Mentorship Gap

Junior students struggle to find guidance due to the lack of a structured mentorship system, leaving them unsure of who to approach. Without a structured platform, seniors get overwhelmed by individual requests and valuable peer knowledge transfer never happens systematically.

2.1.4 Missed Career Opportunities

Alumni who want to recruit from their university have no professional channel to post jobs, making platforms like Facebook feel inappropriate for formal opportunities. There is no way to filter students by skills or qualifications, creating a disconnect between alumni and current students. As a result, students miss valuable job opportunities while alumni and companies lose access to talented graduates.

2.1.5 Club Management Chaos

Student organizations struggle to promote events and track RSVPs across fragmented platforms, with announcements getting buried in social media noise. This leads to low event attendance, heavy administrative burden, and difficulty recruiting members beyond their existing networks.

2.1.6 No University Oversight

Universities relying on external platforms like Facebook and WhatsApp have no way to verify users, moderate content, or maintain an official communication channel with guaranteed reach. This leaves institutions with zero visibility into student engagement data and no control over the information being shared within their own community.

2.2 The Solution: UniConnect

UniConnect addresses these pain points by providing:

✓ Single Platform for all academic networking needs

✓ Skill-Based Discovery to find teammates beyond your immediate circle

✓ Structured Mentorship with scalable group sessions

✓ Professional Career Board for alumni-student connections

✓ Centralized Events & Announcements with priority-based delivery

✓ University-Verified Ecosystem with proper moderation and control

3 USER REQUIREMENT ANALYSIS

3.1 Introduction

This chapter presents an analysis of user requirements for the UniConnect platform. The requirements were gathered through stakeholder interviews and observations of current university communication patterns to ensure the platform addresses genuine needs within the university ecosystem.

3.2 Primary Stakeholder Groups

Stakeholder Type	Role in Platform	Priority
Students	Primary users seeking academic collaboration, mentorship, and career opportunities	High
Alumni	Job/internship providers, industry connection to students	Medium
Club Administrators	Event organizers, community builders, student engagement leaders	Medium
System Administrators	Platform moderators, content managers, technical support	High

Table 1: Primary Stakeholder Groups

3.3 Secondary Stakeholders

- **University Administration:** Interested in student engagement metrics and campus activity monitoring
- **Faculty Members:** Potential users for posting opportunities and monitoring student activities
- **Project Supervisors:** May use platform to track team formation and project progress

3.4 Requirements Gathering Methodology

3.4.1 Stakeholder Interviews (Primary Method)

Semi-structured interviews with 2 students

3.4.2 Observational Studies

- Analysis of current communication channels (WhatsApp groups, Facebook, Email)
- Observation of club recruitment processes
- Review of project team formation methods

3.5 Real-Life Stakeholder Profiles

3.5.1 STAKEHOLDER 1: STUDENT - LAMISA IBNAT ZAMAN

Profile Overview:

Attribute	Details
Name	Lamisa Ibnat Zaman
Department	CSE
Program	B.Sc. in SWE
Year of Study	1st Year
Role	Seeking networking, seeking mentorship

Quote from Lamisa:

"Since most juniors do not know many seniors personally, reaching out to seniors can be difficult. For example, I find it difficult to determine which senior has a good understanding of a specific subject or area. Also, seniors' ability to relate to the problems that juniors face in a certain subject can be beneficial in teaching; therefore, if seniors were to teach examples of topics more simply and offer their support in terms of understanding how to do certain projects, or pass an exam, they would be extremely valuable."

Key Problems Identified:

- **Unknown Senior Expertise:** Cannot identify which seniors have mastered specific subjects or have strong understanding of particular topic areas (Circuits, Calculus, Programming, etc.)
- **No Personal Connection to Seniors:** Most juniors don't personally know seniors, making cold outreach uncomfortable and difficult
- **Missed Peer Learning Value:** Valuable peer-to-peer knowledge transfer (shortcuts, tips, common mistakes, study strategies) is lost due to lack of connection channels

Stated Requirements:
High Priority:

- Searchable mentor directory with senior profiles showing subject expertise
- Group mentorship sessions where multiple juniors can learn together

Medium Priority:

- Reviews/testimonials from past mentees describing how mentor helped them

3.5.2 STAKEHOLDER 2: STUDENT - LABIBAH HOQUE

Profile Overview:

Attribute	Details
Name	Labibah Haque
Department	CSE
Program	B.Sc. in SWE
Year of Study	4th Year
Role	Club member, seeking internships

Quote from Labibah:

"Sometimes it becomes quite difficult when you want to find an alumni in the company where you are applying for job. Having a networking site where alumni post jobs/internships from their company will be a great help in this case. And at present searching for club events requires searching for the club fb pages and keeping tabs on them regularly. If we don't follow it regularly the event post may be lost or we may never find it which is actually quite difficult."

Key Problems Identified:

- No central alumni platform where alumni can post available jobs and students can view them
- Club event announcements get lost in social media noise

Stated Requirements:
High Priority:

- Job board with job availability posting

Medium Priority:

- Event calendar for club activities
- Connection with alumni for career guidance

3.6 Observational Studies

3.6.1 Observation 1: Authentication and Verified Access

Key Points:

- The platform should be limited to university users
- Complex ID verification or university database integration is not part of the final scope

Key Problems:

- Open registration may allow fake users
- University database integration would make development more complex
- Manual ID verification would take extra time
- Users need a simple and secure registration process

Requirements:

- University email-only registration
- Email domain validation
- Email verification using verification link
- Login with email and password
- JWT-based authentication
- Password reset

3.6.2 Observation 2: User Profile and Privacy

Key Points:

- Different user types need different profile information
- Students need skills and interests, alumni need professional information, faculty need academic information, and clubs need organizational profiles

Key Problems:

- Users do not have a unified university networking identity
- Students cannot discover people by skills or interests
- Users need control over what information is visible
- Profile photos and tags improve discoverability

Requirements:

- Multi-type user profiles
- Profile creation and editing

- Profile photo upload
- Skills and interests tagging
- Privacy settings
- Role-based profile fields

3.6.3 Observation 3: Project Teammate Finder

Key Points:

- Students often struggle to find suitable teammates outside their known circle
- Skill-based and graph-based matching can improve team formation

Key Problems:

- Students rely on friends for team formation
- Skill discovery is difficult
- Project posts are not centralized
- Team management is informal

Requirements:

- Create project posts
- Browse project posts with department filters
- Apply to projects
- Team formation and management
- Teammate matching using graph-based similarity algorithm

3.6.4 Observation 4: Peer Mentorship

Key Points:

- The mentorship system will be student-to-student only
- Alumni career mentorship is removed from the final scope

Key Problems:

- Juniors cannot identify suitable senior mentors
- Senior students cannot manage mentorship requests easily
- Mentorship lacks tracking
- Career mentorship with alumni would increase complexity

Requirements:

- Senior students register as mentors
- Junior students search for mentors
- Students send mentorship requests
- Mentors accept or reject requests
- Active mentorship tracking
- Chat between accepted mentor and mentee

3.6.5 Observation 5: Club Management

Key Points:

- Club activities are scattered across social media and messaging apps
- A centralized club directory and management system can improve engagement

Key Problems:

- Students miss club updates
- No central club directory exists
- Club membership is manually managed

Requirements:

- Club directory
- Club profiles
- Join and leave clubs
- Membership management
- Club-specific features

3.6.6 Observation 6: Events and Announcements

Key Points:

- Events and announcements are important for campus engagement
- They often get lost in social media noise

Key Problems:

- Event posts are scattered
- Students miss registration deadlines
- No centralized calendar view exists
- Important announcements are not targeted

Requirements:

- Create, edit, and delete events
- Event calendar view
- RSVP system
- Event reminders
- Targeted announcements
- Announcement feed

3.6.7 Observation 7: Career Features

Key Points:

- The career feature is simplified to a job/internship board only
- Alumni career mentorship and career resources are removed or deferred

Key Problems:

- Students miss job and internship posts
- Opportunities are scattered across groups

Requirements:

- Job and internship board
- Alumni can post jobs
- Students can browse jobs with filters
- Students can view job details and requirements
- Students can apply via external link or email

3.6.8 Observation 8: One-on-One Chat

Key Points:

- Users need direct communication after forming connections, mentorships, club relations, or project teams

Key Problems:

- Communication is fragmented across Messenger, WhatsApp, and email
- Important project, mentorship, or club messages may be lost

Requirements:

- Real-time messaging using WebSocket

- Conversation list
- Message history
- Message notifications
- Chat access based on connection or relevant relationship

4 FLOW CHART

4.1 Authentication & Authorization

4.2 Project Teammate Finder

4.3 Peer-to-Peer Mentorship

4.4 One-to-One Chat

4.5 Events & Announcements

4.6 Career Board

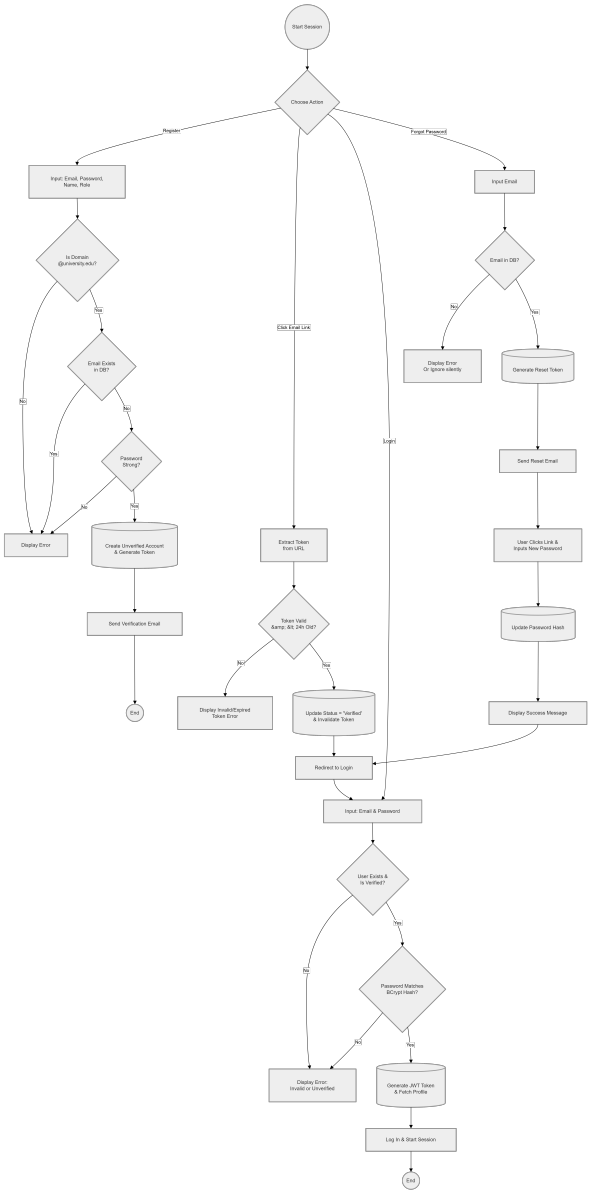
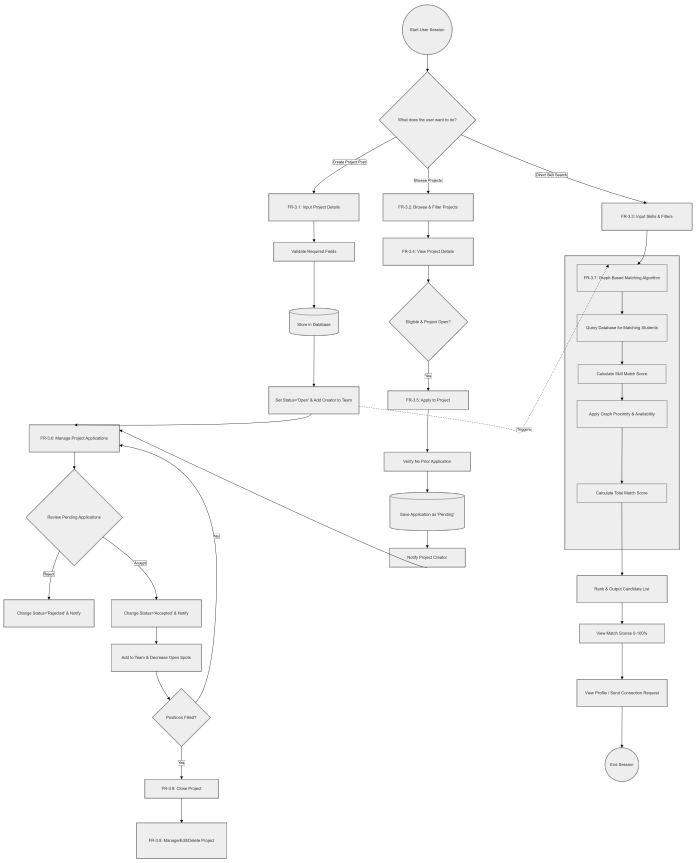
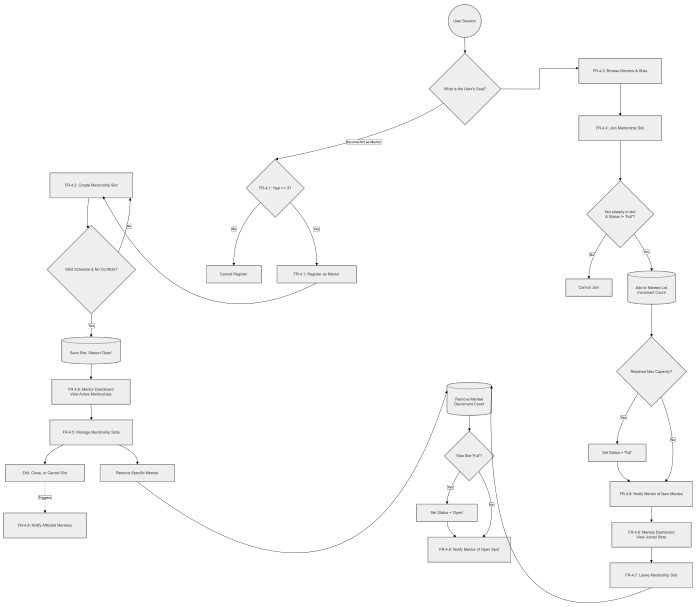


Figure 1: Authentication & Authorization Flowchart



2

Figure 2: Project Teammate Finder Flowchart



3

Figure 3: Peer-to-Peer Mentorship Flowchart

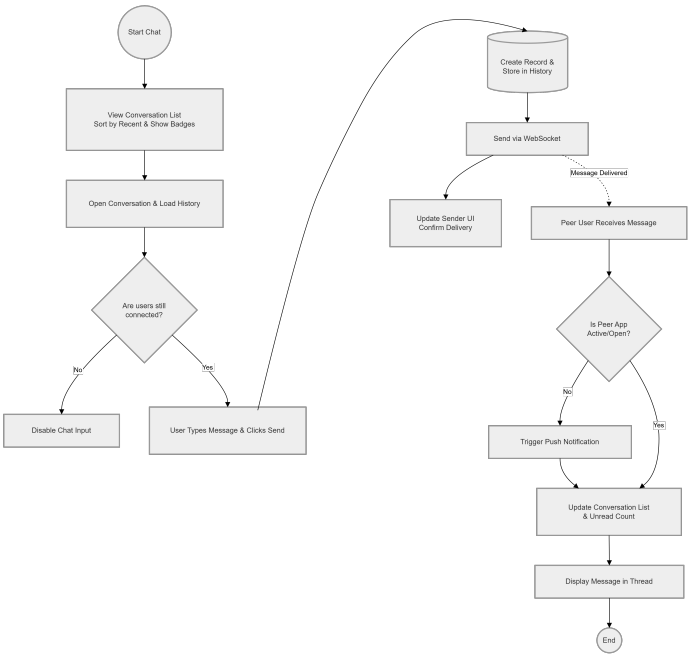
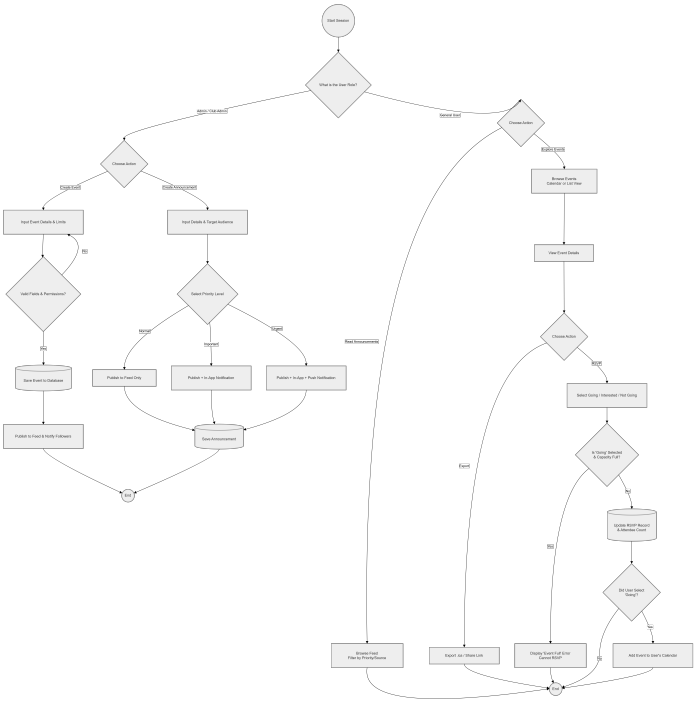
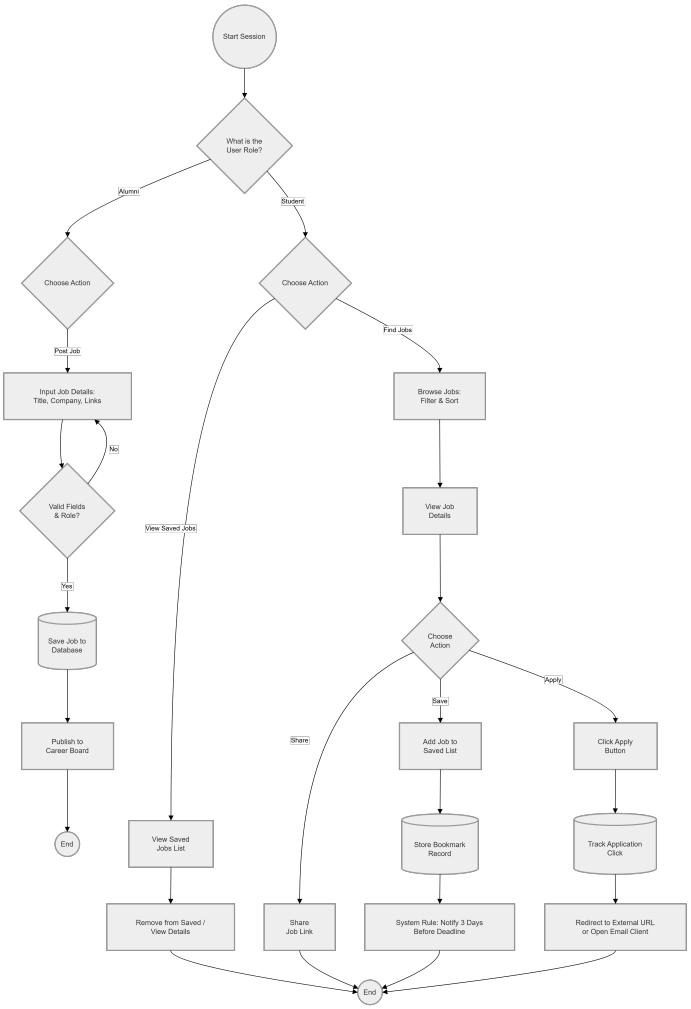


Figure 4: One-to-One Chat Flowchart



2

Figure 5: Events & Announcements Flowchart



3

Figure 6: Career Board Flowchart

5 KEY FEATURES AND USE CASE DESCRIPTIONS

5.1 Overview of Features

1. Authentication & Authorization
2. Project Teammate Finder
3. Peer-to-Peer Mentorship
4. One-to-One Chat
5. Events & Announcements
6. Career Board

5.2 Feature 1: Authentication & Authorization

5.2.1 Feature Overview

The authentication system ensures that only verified university members can access UniConnect. It provides secure login, email verification, and role-based access control.

5.2.2 USE CASE 1.1: User Registration

Use Case ID: UC-AUTH-01
Use Case Name: Register New User Account
Actor: Unregistered User (Student/Alumni)
Preconditions:

- User has a valid university email address
- User is not already registered

Postconditions:

- User account created with status "Unverified"
- Verification email sent
- User cannot login until email is verified

5.2.3 USE CASE 1.2: Email Verification

Use Case ID: UC-AUTH-02
Use Case Name: Verify Email Address
Actor: Unregistered User
Preconditions:

- User has completed registration (UC-AUTH-01)
- User has received verification email
- Verification token is not expired (¡ 24 hours old)

Postconditions:

- User account status changed to "Verified"
- User can now login to the system

5.2.4 USE CASE 1.3: User Login

Use Case ID: UC-AUTH-03
Use Case Name: Login to UniConnect
Actor: Registered User (Student/Alumni/Club Admin/System Admin)
Preconditions:

- User has registered account
- User email is verified
- User account is active (not suspended)

Postconditions:

- User is authenticated and logged in
- JWT token generated and stored
- User session is active
- User can access role-appropriate features

5.3 Feature 2: Project Teammate Finder

5.3.1 Feature Overview

The Project Teammate Finder helps students discover and recruit team members for academic projects. It includes two primary modes: creating project posts and directly searching for students by skills.

5.3.2 USE CASE 2.1: Search Teammates by Skills (Direct Search)

Use Case ID: UC-PROJ-01
Use Case Name: Search for Teammates by Required Skills
Actor: Student
Preconditions:

- User is logged in as a Student
- User profile includes skills information

Postconditions:

- Search results displayed with match scores
- User can contact potential teammates

5.3.3 USE CASE 2.2: Create Project Post

Use Case ID: UC-PROJ-02
Use Case Name: Create Project Post to Find Teammates
Actor: Student
Preconditions:

- User is logged in as a Student
- User has verified email

Postconditions:

- Project is created and visible to all students
- Project status is "Open"
- Creator is listed as team member
- Project appears in browse/search results
- Matching algorithm identifies potential teammates

5.3.4 USE CASE 2.3: Apply to Project

Use Case ID: UC-PROJ-03
Use Case Name: Apply to Join a Project
Actor: Student
Preconditions:

- User is logged in as a Student
- Project exists and is in "Open" status
- User is not already a team member
- User has not already applied

Postconditions:

- Application created with status "Pending"
- Project creator receives notification
- Application visible in applicant's dashboard
- Application can be viewed by project creator

5.3.5 USE CASE 2.4: Manage Project Applications

Use Case ID: UC-PROJ-04
Use Case Name: Review and Respond to Project Applications
Actor: Student (Project Creator)
Preconditions:

- User is logged in as a Student
- User is the creator of the project
- Project has received applications

Postconditions:

- Application status updated (Accepted/Rejected)
- If accepted: Applicant added to team, position filled
- Notifications sent to applicant
- Project may be closed if all positions filled

5.4 Feature 3: Peer-to-Peer Mentorship

5.4.1 Feature Overview

The Peer-to-Peer Mentorship system connects junior students with senior students through structured, slot-based group mentorship sessions. Mentors create scheduled time slots (e.g., "Calculus Help - Mondays 5 PM") and multiple mentees (7-10) can join each slot.

5.4.2 USE CASE 3.1: Register as Mentor

Use Case ID: UC-MENTOR-01
Use Case Name: Register as Peer Mentor
Actor: Student (Year 2+)
Preconditions:

- User is logged in as a Student
- User's year of study is 2 or higher
- User is not already registered as a mentor

Postconditions:

- User is registered as a mentor
- Mentor profile created and visible
- User can now create mentorship slots
- User appears in mentor directory

5.4.3 USE CASE 3.2: Create Mentorship Slot

Use Case ID: UC-MENTOR-02
Use Case Name: Create Scheduled Mentorship Slot
Actor: Student (Registered Mentor)
Preconditions:

- User is logged in as a Student
- User is registered as a Mentor (UC-MENTOR-01 completed)

Postconditions:

- Mentorship slot created and visible in directory
- Slot status is "Open"
- Students can browse and join slot
- Mentor dashboard shows new slot

5.4.4 USE CASE 3.3: Join Mentorship Slot

Use Case ID: UC-MENTOR-03
Use Case Name: Join a Mentorship Slot
Actor: Student (Mentee)
Preconditions:

- User is logged in as a Student
- Mentorship slots exist in the system
- User is not already in the slot

Postconditions:

- User enrolled in mentorship slot
- Mentee count incremented
- Slot may change to "Full" status
- Notifications sent to mentor and mentee
- Calendar/schedule updated for student
- Meeting details shared with student

5.4.5 USE CASE 3.4: Manage Mentorship Slot

Use Case ID: UC-MENTOR-04
Use Case Name: Manage and Modify Mentorship Slot
Actor: Student (Mentor)
Preconditions:

- User is logged in as a Student
- User is a registered mentor
- User has created at least one mentorship slot

Postconditions:

- Slot details updated (if edited)
- Mentees notified of changes
- Slot status changed (if closed/cancelled)
- Mentee count adjusted (if mentee removed)

5.5 Feature 4: One-to-One Chat

5.5.1 Feature Overview

The One-to-One Chat system enables real-time messaging between first-degree connections using WebSocket technology. Users can send text messages, view conversation history, and receive instant notifications.

5.5.2 USE CASE 4.1: Send Message

Use Case ID: UC-CHAT-01
Use Case Name: Send Real-Time Text Message
Actor: Student/Alumni/Club Admin
Preconditions:

- User is logged in
- User and recipient are 1st-degree connections
- WebSocket connection is active

Postconditions:

- Message sent and stored in database
- Message appears in both users' conversation history
- Recipient receives notification (if appropriate)
- Conversation moved to top of conversation list

5.5.3 USE CASE 4.2: View Conversation History

Use Case ID: UC-CHAT-02
Use Case Name: View Complete Message History with Connection
Actor: Student/Alumni/Club Admin
Preconditions:

- User is logged in
- User has at least one conversation

Postconditions:

- Complete conversation history displayed
- Unread messages marked as read
- User can send new messages from this view
- Conversation list updated with new read status

5.6 Feature 5: Events & Announcements

5.6.1 Feature Overview

The Events & Announcements feature allows Club Admins and System Admins to create campus events with RSVP tracking and targeted announcements with priority levels.

5.6.2 USE CASE 5.1: Create Event

Use Case ID: UC-EVENT-01
Use Case Name: Create Campus Event with RSVP
Actor: Club Admin, System Admin
Preconditions:

- User is logged in as Club Admin or System Admin
- User has permission to create events

Postconditions:

- Event created and visible to all users
- Event appears in event calendar and feed
- Users can RSVP to event
- Notifications sent to relevant users

5.6.3 USE CASE 5.2: RSVP to Event

Use Case ID: UC-EVENT-02
Use Case Name: Respond to Event Invitation
Actor: Student, Alumni, Club Admin
Preconditions:

- User is logged in
- Event exists and is upcoming
- RSVP is enabled for event

Postconditions:

- User's RSVP recorded
- Attendee count updated
- Event added to user's calendar (if "Going")
- Reminder notification scheduled
- Organizer can see RSVP in attendee list

5.6.4 USE CASE 5.3: Create Announcement

Use Case ID: UC-ANNOUNCE-01
Use Case Name: Create Priority-Based Announcement
Actor: Club Admin, System Admin
Preconditions:

- User is logged in as Club Admin or System Admin
- User has permission to create announcements

Postconditions:

- Announcement published to feed
- Notifications sent based on priority
- Target audience receives announcement
- Announcement visible until expiry (if set)

5.7 Feature 6: Career Board

5.7.1 Feature Overview

The Career Board allows Alumni to post job and internship opportunities, and Students to browse, filter, save, and apply to positions based on their skills and interests.

5.7.2 USE CASE 6.1: Post Job Opportunity

Use Case ID: UC-CAREER-01
Use Case Name: Post Job or Internship Opportunity
Actor: Alumni
Preconditions:

- User is logged in as Alumni
- User has verified profile

Postconditions:

- Job posted and visible to all students
- Students can browse, filter, and save job

5.7.3 USE CASE 6.2: Browse and Apply to Jobs

Use Case ID: UC-CAREER-02
Use Case Name: Browse Jobs and Apply
Actor: Student
Preconditions:

- User is logged in as Student
- Job postings exist in system

Postconditions:

- User has viewed job details
- Job may be saved to user's list
- Application tracked (if applied)
- Reminder scheduled (if saved with deadline)

6 TOOLS AND TECHNOLOGY

6.1 Tech Stack

Layer	Tools
Frontend	Flutter
Backend	Java Spring Boot
Database	PostgreSQL 14+
Real-time Communication	WebSocket
Authentication	JWT + BCrypt

Table 2: Technology Stack

6.2 Development Tools

Tool	Purpose
IntelliJ IDEA	Backend (Java) development
VS Code + Android Studio	Flutter development
GitHub	Version Control
Postman + Swagger	API Testing
Trello	Project management and issue tracker

Table 3: Development Tools

6.3 Project Structure

Aspect	Approach
Architecture	Modular OOP
SDLC Model	Agile Model
Design Principles	SOLID Principles and Clean Code

Table 4: Project Structure and Methodology

7 PROPOSED TIMELINE

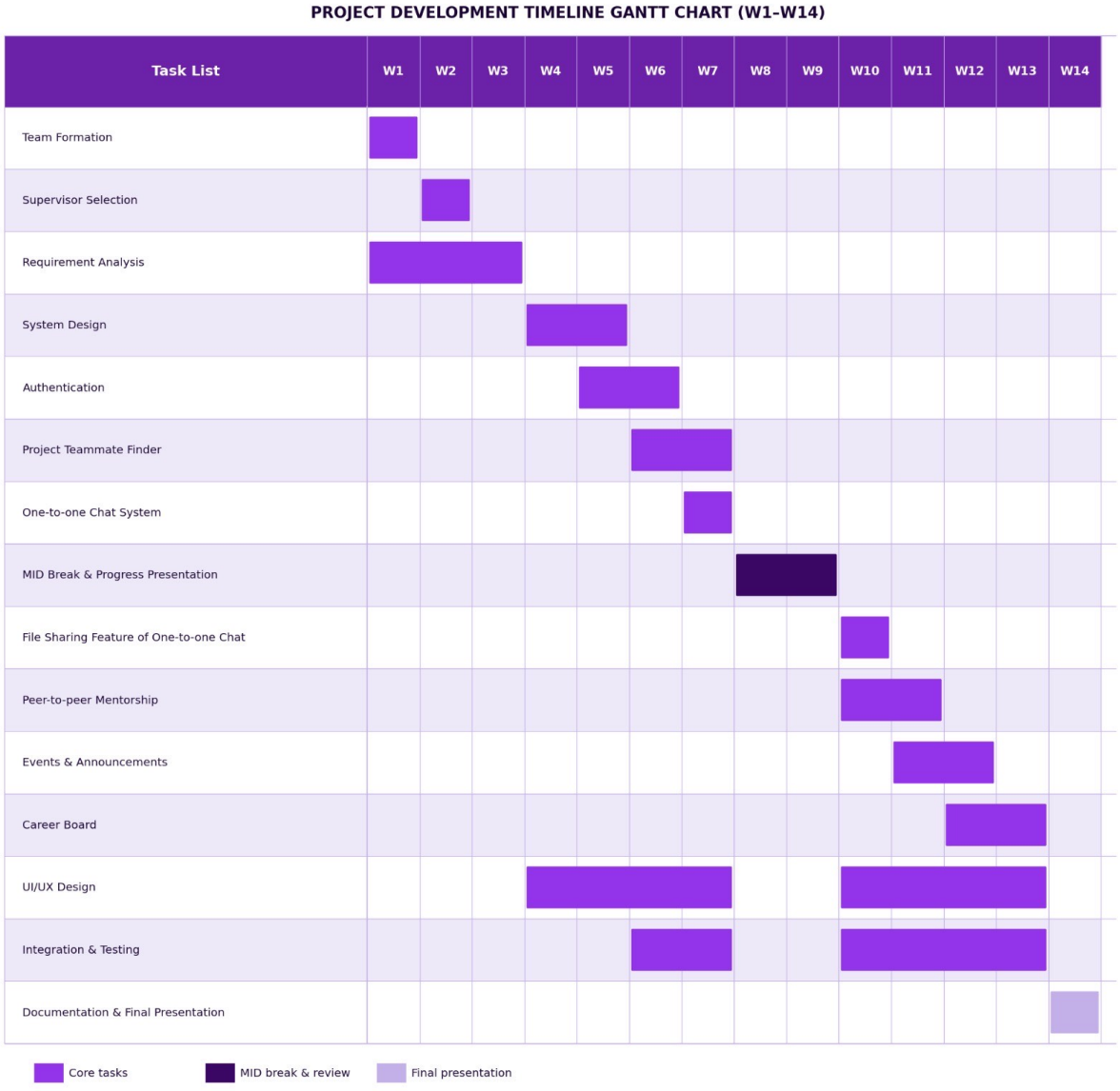


Figure 7: Proposed Timeline

8 ACTION TAKEN BASED ON FEEDBACK

8.1 Feedback 1: Direct Skill-Based Search for Teammates

8.1.1 Stakeholder Feedback

During the Project Proposal Presentation, a critical limitation was identified in the initial Project Teammate Finder design:

Original Feedback:

"I get the idea where you mentioned that project title and description should first be created, but a lot of times, teammates search with specific skills- they first form a team and then decide which project to work on."

8.1.2 Problem Identified

The initial design only allowed students to find teammates through project posts, requiring them to:

- Create a formal project post first
- Wait for applications
- Be committed to a specific project scope

This workflow did not support exploratory teammate discovery or informal collaboration inquiries.

8.1.3 Action Taken

In response to this feedback, we implemented **USE CASE 2.1: Search Teammates by Skills (Direct Search)** as a standalone feature in the Project Teammate Finder module.

Key Implementation Details:

1. **Skill-Based Search Interface:**
   - Students can enter required skills (e.g., "Python", "Machine Learning", "React")
   - Filter by department, year of study, and availability
   - No need to create a project post first
2. **Match Scoring Algorithm:**
   - Graph-based similarity algorithm calculates match scores
   - Results ranked by skill match percentage
   - Shows students with highest relevance first
3. **Direct Contact:**
   - Students can view detailed profiles of matched users
   - Initiate direct chat conversations
   - Send connection requests

8.1.4 Impact and Benefits

This feature addition provides:

- **Flexibility:** Students can explore potential collaborators before committing to a formal project
- **Faster Discovery:** Immediate access to skill-matched students without waiting for applications
- **Informal Collaboration:** Supports study groups, hackathon teams, and informal projects
- **Reduced Barrier:** No need to write formal project descriptions for exploratory searches

8.1.5 Workflow Comparison

Before (Project-Post Only):

1. Student creates detailed project post
2. Waits for applications
3. Reviews applicants
4. Accepts/rejects based on profiles

After (With Direct Search):

1. Student searches by skills (e.g., "Flutter", "UI/UX")
2. Views ranked list of matching students
3. Contacts relevant students directly
4. Discusses collaboration informally before formalizing